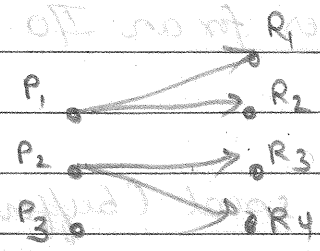
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Q: A system has 3 processes sharing 4 resources. If each process need a max of 2 units, then

(a) deadlock can never occur
(b) deadlock may occur
(c) deadlock has to occur
(d) none.



Q: An O.S contain 3 user processes each requiring 2 units of resource R. Min no. of units of R such that no deadlock will ever occur.

(i) 5 (ii) 2
(iii) 3 (iv) 4

XML

XML file + DTD = XML Document System.

+ CSS / XSL

Structure : XML document is similar to tree structure with single root element and one or many child element. child element of same level are referred to as siblings.

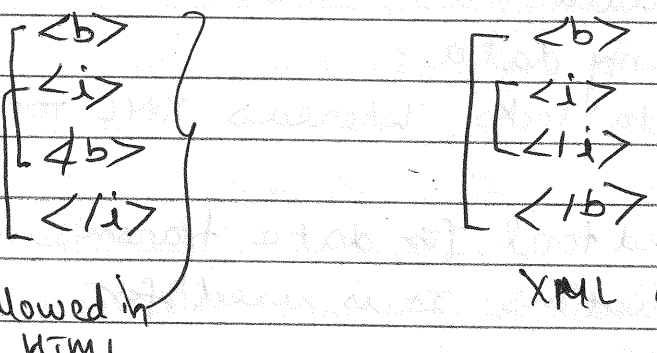
Tags

(i) All XML tags must have closing tag unlike HTML

(ii) XML tags are case sensitive.

Nesting

XML is very strict with nesting rules.



allowed in HTML but not in XML.

XML only follows this

XML attributes

XML attributes values must be double quoted / or single quoted. else will be considered error.

Comments in XML

<!-- -->

White spaces

XML doesnot change original text. It lets white spaces as they are.

eg: I am kiran.

XML uses XSL to describe data.

Dt.: Pg.: Delta

New line in XML → new line is stored as LF.

Naming rules

(i) cannot start with number or punctuation character

(ii) cannot " " letters like XML.

(iii) " " spaces.

(iv) names can have letters, numbers

(v) no reserved words are there.

(vi) avoid using (:).

Difference b/w Attributes and Elements

(i) It is better to write an attribute as an element since elements are expandable

(ii) Attributes cannot have tree-like structure unlike elements

(iii) Attributes cannot have multiple values whereas an element can have.

XML validation

There are XML validator softwares which can carry out syntax checks for XML. This validation is important because an XML application stops when an error is encountered in XML document, whereas if an error is there in HTML code, it is displayed. A valid XML document obeys rule of DTD (Document Type Definition)

Well formed XML Document

An XML document having no syntax error is called well-formed document.

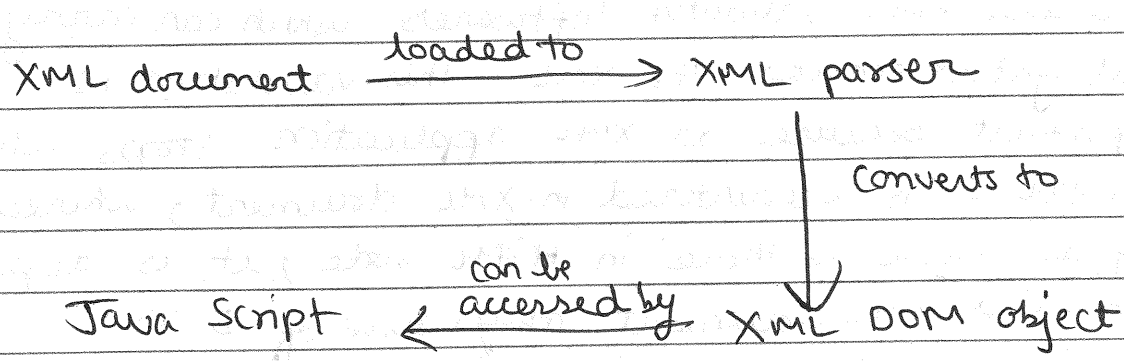
The syntax rules are

(i) Every XML document should have a root element
(ii) All elements have closing tags
(iii) All tags are case sensitive
(iv) Attributes must have quoted values
(v) Nesting of elements must not overlap

XML parser

A parser is a program which recieves an input in form of sequential program and breaks into several parts which are then managed by other program or applications.

The parser ~~breaks~~ converts XML into Javascript accessible object (XML DOM — XML Data Object Module). A DOM view XML document as tree structure with elements as nodes. Most browsers have inbuilt XML parser.



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XML Declaration.

used to indicate specified XML document.
It is first line of document.

```
<? xml version = "1.0" encoding = "UTF-8"
  standalone = "no" ? >
```

Version : identifies version of XML. Recommended '1.0'
encoding : specifies character set (optional). By default : UTF-8
standalone : whether document needs external entities or not.
'Yes' specifies XML document has DTD.
'NO' specifies XML document is linked to external entity. By default: 'Yes'.

(DOCTYPE) Declaration

↳ Document Type

DTD are used in XML to specify information about a document, including document's root element and Data Type Definition.

The doctype (document type) declaration begins with <! DOCTYPE> declaration.

Syntax:

```
<! DOCTYPE dtd_name SYSTEM = "dtd-file_name.dtd">
```

Simplest DOCTYPE declaration identifies only root element of document

```
<! DOCTYPE rootelement >
```

i) Root / Document Element

Each XML document has exactly one single root element. Root Elements are also called parent element.

e.g.

```
<School>
<Student>
<Roll_No> 12 </Roll_No>
</Student>
</School>
```

ii) Elements

Elements are the basic units used to identify and store data and are represented by using tags. Elements are divided into two basic categories empty and nested (child):

a) Empty Elements: If an element does not have any content, it is called an empty element.

e.g.

```
<Product>
<Item item_id="24" />   } empty element
</Product>
```

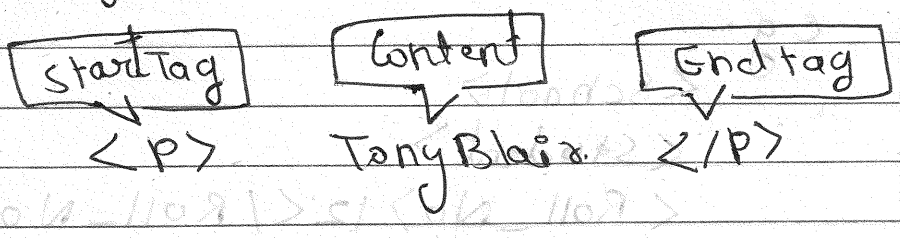
b) child Elements: A child elements is a nested element, i.e, element inside another element.

e.g.

```
<book>                          Parent element
<title> Harry Potter </title>   } child elements
<author> J.K. Rowling </author> }
</book>
```

iii) Tags

Tags are used to identify elements in an XML document. XML tags consist of opening and closing angular brackets (< >).

e.g. 

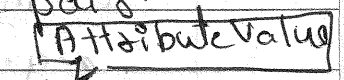
a) Start tag
eg: <tag>

b) End tag
e.g: </tag>

c) Empty tag
e.g: <tag/>

iv) Attributes

An attribute specifies a single property of an element, using a name-value pair.



e.g. <PRODUCT PRODUCTID="1" CATEGORY

v) Content

e.g. <PRODUCTNAME>Books</PRODUCTNAME>

Content of <PRODUCTNAME> element

vi) Special Characters

Special characters are also known as pre-defined entities

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character	Predefined Entity
&	&
<	<
>	>
"	"
'	'

→ Element Occurence Indicators

The plus (+), star (*) and question mark (?) characters in the element content are occurence indicators. They indicate whether and how element in the child list can repeat.

i) An element followed by no occurence indicator must appear once and only once in the element being defined

ii) An element followed by a "+" character must appear one or several times in the element being defined. The element can repeat.

iii) An element followed by a "*" character can appear once or not at all in the element being defined.